

Masters Project Research Log

Masters in Electronic and Computer Engineering 2024/2025

Student Name: Katherina Keogh

Student ID: 19423752

Project Title: RFID Side Channel Analysis for IoT Devices

Please read before making entries in this log

The purpose of this Project Research Log is to capture concise, focused summaries of research materials you read, as you progress through your project. The emphasis is to record (i) how the material you have read will determine or influence your project solution approach and (ii) your assessment of the key strengths and weaknesses of the solutions, methods, technologies, etc. proposed in the material you have read.

In the first stage of your project, the literature review, use the Log to capture this information for the key papers you have read (for example, the three most important papers of your 10 literature review references). As your project progresses into the design and implementation phases, you will need to continue to search the literature so you can review, revise and refine your initial thinking and the details of your approach to a project solution. Use this Research Log to capture your continued research reading and its influence on your project design and implementation.

Be selective about what you record in this log. Do not use it as an informal notebook while you are reading a new paper. Only make an entry after you have read a paper that you consider important to the development of your project solution. It is expected that, by the end of the project, you will have made **between 10 and 20 entries (20 maximum)**.

Share your log with your supervisor for viewing throughout the project. You will submit the final version of the log for grading, at the end of the project implementation period. It will be assessed on the basis of how well you have used your analysis of the literature to inform your project design, implementation and the evaluation of your project results. The Research Log contributes **5%** to the overall project mark.

Note: All entries you make in this log must use the prescribed format shown on the next page. You will maintain other notes as you progress through your project but they should not be recorded here. Fill in the details where the *** signs are.

Statement of project problem / research question (maximum 200 words)

This statement should be periodically reviewed and updated, as necessary, as your project progresses and you gain further insight into the detailed project challenges, requirements and objectives as your project work moves from background reading, literature review, initial project design planning and detailed design and implementation. Initially, start by stating your current understanding of the project objectives. After each meeting with your supervisor, review and refine your project problem statement, as required.

The main objective of this project is to investigate whether MIFARE Classic RFID cards, widely used for access control and payment systems, leak key dependent information through physical side channels such as power consumption and electromagnetic (EM) emissions during authentication. While previous research has exposed weaknesses in the Crypto1 cipher and demonstrated protocol based attacks, there is limited evidence on the extent to which physical signals can be exploited for practical key recovery. The project aims to collect, process, and statistically analyse side channel measurements from repeated authentication attempts using different keys, assessing correlations between recorded signals and key properties like Hamming weight.

1. Side-Channel Analysis of Cryptographic RFIDs with Analog Demodulation**A complete reference for the paper**

Kasper, T., Oswald, D., & Paar, C. (2012). *Side-Channel Analysis of Cryptographic RFIDs with Analog Demodulation* (pp. 61–77). https://doi.org/10.1007/978-3-642-25286-0_5

Summary of paper (maximum 100 words)

This paper presents an analog demodulator that boosts the clarity of electromagnetic side-channel signals from RFID tags during cryptographic operations. Plugged into standard lab equipment, it filters and amplifies leaked EM emissions so that secret keys can be extracted more reliably from commercial cards (MIFARE DESFire). The authors demonstrate that, with this low cost add on, key recovery attacks succeed using modest hardware, revealing that even secure RFID implementations are vulnerable in practice.

How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)

The demodulator's role in improving EM signal quality directly guides my experimental setup for MIFARE Classic cards. It shows how hardware enhancements can make leakage easier to detect, informing my choice of probe placement, signal conditioning, and the importance of maximising trace clarity before statistical analysis.

What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)**Strengths:**

- Simple, low cost device that significantly improves EM trace quality
- Validated on real world RFID cards, making attacks practical
- Easy to integrate into existing lab setups

Weaknesses:

- Focuses on signal capture, not on preventing leaks
- Tested on limited devices, performance on others may vary

- Requires precise calibration and tuning for optimal results

2. Analysis of Vulnerabilities in Radio Frequency Identification (RFID) Systems.

A complete reference for the paper

Jain, R., Kumar Chaudhary, D., & Kumar, S. (2018). Analysis of Vulnerabilities in Radio Frequency Identification (RFID) Systems. *2018 8th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, 453–457.
<https://doi.org/10.1109/CONFLUENCE.2018.8442623>

Summary of paper (maximum 100 words)

This paper analyses practical vulnerabilities in RFID systems by demonstrating attacks such as cloning, data tampering, and denial-of-service on university ID cards. Through experiments using common tools, the authors show how easily tags can be copied or disabled, exposing risks to access control and identity applications. They provide detailed attack procedures and emphasise the need for stronger authentication and data integrity measures in RFID systems.

How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)

This paper broadens my understanding of RFID security threats beyond side channels, highlighting how physical and protocol weaknesses can be exploited. It reinforced the importance of examining all attack surfaces, motivating my focus on side channel leakage as an additional vulnerability. The experimental approach to realistic attacks influenced my design of controlled test scenarios with Proxmark3 and oscilloscope measurements.

What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)

Strengths:

- Hands on experiments on real RFID cards
- Clear demonstration of multiple attack types
- Uses readily available tools, making findings accessible

Weaknesses:

- Limited focus on countermeasures or defences
- Small scale tests may not generalize to all RFID products
- Does not include statistical analysis or quantitative assessment of attack success rates

3. Introduction to differential power analysis. *Journal of Cryptographic Engineering*

A complete reference for the paper

Kocher, P., Jaffe, J., Jun, B., & Rohatgi, P. (2011). Introduction to differential power analysis. *Journal of Cryptographic Engineering*, 1(1), 5–27.
<https://doi.org/10.1007/s13389-011-0006-y>

Summary of paper (maximum 100 words)

This paper introduces Differential Power Analysis (DPA), a technique that exploits variations in a device's power consumption during cryptographic operations to recover

secret keys. By collecting many power traces and using statistical correlation with hypothetical power models (like Hamming weight), the authors demonstrate successful key extraction attacks on DES and AES smartcards. This work proves that even well designed ciphers leak information through their physical power usage.

How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)

DPA is the foundation of my analysis. It explains how to link side channel traces to key dependent values. This paper guided my choice of Hamming weight leakage models and statistical methods (t-tests, Pearson correlation) for detecting key information in power and EM measurements from MIFARE Classic cards.

What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)

Strengths:

- Provides a clear, generalisable method for key recovery via power analysis
- Validated on real DES/AES implementations
- Lays groundwork for many side channel techniques

Weaknesses:

- Requires thousands of traces for reliable results
- Assumes precise trace alignment and low noise
- Focuses on power, not EM, so additional adaptations are needed for EM leakage scenario

4. Side-channel analysis and machine learning: A practical perspective

A complete reference for the paper

Picek, S., Heuser, A., Jovic, A., Ludwig, S. A., Guilley, S., Jakobovic, D., & Mentens, N. (2017). Side-channel analysis and machine learning: A practical perspective. *2017 International Joint Conference on Neural Networks (IJCNN)*, 4095–4102. <https://doi.org/10.1109/IJCNN.2017.7966373>

Summary of paper (maximum 100 words)

This paper examines applying machine learning to side channel analysis, comparing algorithms like Support Vector Machines and Random Forests against traditional template attacks. By extracting statistical features from power traces and training classifiers, the authors show ML methods can more accurately distinguish key dependent patterns, even with noisy data or fewer traces. Experiments on AES implementations demonstrate ML based attacks often succeed with fewer samples and show a greater ability to measurement variability.

How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)

The paper's demonstration of ML techniques offers an alternative to classical statistical methods for analysing side channel data. It suggests that, after collecting EM or power traces from MIFARE Classic cards, training a classifier could improve key recovery accuracy compared to correlation analysis alone. This informs potential future enhancements to my analysis.

What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)

Strengths:

- Achieves higher accuracy with noisy or limited data
- Requires fewer traces than classical attacks
- Clearly outlines feature extraction and classifier training steps

Weaknesses:

- Depends on labelled profiling data, which may be unavailable
- Risk of overfitting models to specific devices
- Introduces computational complexity and parameter tuning challenges

5. Side-Channel Attack on a Protected RFID Card

A complete reference for the paper

Xu, R., Zhu, L., Wang, A., Du, X., Choo, K.-K. R., Zhang, G., & Gai, K. (2018). Side-Channel Attack on a Protected RFID Card. *IEEE Access*, 6, 58395–58404.
<https://doi.org/10.1109/ACCESS.2018.2870663>

Summary of paper (maximum 100 words)

This paper demonstrates a successful side channel attack on a commercially protected RFID card implementing DES. By capturing power traces during authentication and applying DPA, the authors recover secret keys despite built in hardware countermeasures. Their experiments on three card models show key extraction in under 5,000 traces, proving that standard protections (noise filtering, random delays) can be bypassed with optimised statistical techniques.

How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)

Their methodology directly informs my approach to measuring and analysing leakage from MIFARE Classic cards. It shows that even counter measured devices leak exploitable information, guiding my choice of feature extraction (peak amplitudes) and statistical tests. The paper's experimental design also helps refine my trace alignment and preprocessing steps to maximize signal clarity.

What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)

Strengths:

- Validates attack possibility on protected, real world RFID cards
- Uses optimised statistical distinguishers to reduce required traces
- Provides detailed procedural guidance for trace capture and analysis

Weaknesses:

- Focuses only on power side channels, not EM emissions
- Requires specialised statistical tools that may be complex to implement
- Experiments limited to DES-based cards, results may differ on other ciphers

6. *Proxmark3 -- Iceman repo.*

A complete reference for the paper
Herrmann, C., Teuwen P., Moiseenko M., Walker M., & Others. (n.d.). <i>Proxmark3 -- Iceman repo</i> . GitHub.
Summary of paper (maximum 100 words)
The Proxmark3 Iceman firmware repository offers enhanced tools and scripts for interacting with RFID tags. It includes commands for card detection, authentication, reading/writing memory blocks, and advanced side channel measurement triggers. The firmware's modular design and active community support enable researchers to script automated RFID operations, capture precise timing events, and integrate with oscilloscopes for waveform collection.
How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)
It was the firmware that was running on the Proxmark
What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)
N/A

7. *MIFARE Classic EV1 1K*

A complete reference for the paper
NXP Semiconductors. (2018). <i>MIFARE Classic EV1 1K - Mainstream contactless smart card IC for fast and easy solution development</i> . NXP. https://www.nxp.com/docs/en/data-sheet/MF1S50YYX_V1.pdf
Summary of paper (maximum 100 words)
The documentation for the MIFARE Classic card
How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)
This documentation is essential for understanding how MIFARE Classic cards authenticate and store data
What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)
N/A

8. *Mifare Tag Ops.*

A complete reference for the paper
<i>Mifare Tag Ops.</i> (2017, August). https://github.com/Proxmark/proxmark3/wiki/Mifare-Tag-Ops#command-set-hf-mf

Summary of paper (maximum 100 words)
The documentation for Proxmark3 commands for the MIFARE Classic card
How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)
This documentation is essential for understanding the types of commands for MIFARE Classic cards
What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)
N/A

9. *Practical Statistics for Data Scientists*

A complete reference for the paper
Bruce Peter, B. A. G. P. (2020). <i>Practical Statistics for Data Scientists, 2nd Edition</i> (2nd ed.). O'Reilly Media, Inc.
Summary of paper (maximum 100 words)
This book provides a practical guide to essential statistical techniques for data analysis, covering topics like hypothesis testing, regression, classification, clustering, and resampling methods.
How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)
The book guided my selection and implementation of statistical tests (t-tests, Cohen's d, Pearson's r), ensuring I apply them correctly and interpret results properly. Its clear explanations of effect sizes and correlation analyses helped me choose appropriate thresholds and report findings.
What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)
Strengths: • Accessible explanations with practical examples in code • Broad coverage of core techniques relevant to data science • Emphasises interpretation and best practices for reproducible analysis Weaknesses: • Not focused on side channel or security specific statistics • Examples use generic datasets rather than hardware leakage traces • Limited depth on advanced topics like high dimensional or time series analysis

10. *A Practical Attack on the MIFARE Classic*

A complete reference for the paper
de Koning Gans, G., Hoepman, J.-H., & Garcia, F. D. (2008). <i>A Practical Attack on the MIFARE Classic</i> (pp. 267–282). https://doi.org/10.1007/978-3-540-85893-5_20

Summary of paper (maximum 100 words)

This paper describes a practical attack on MIFARE Classic cards, detailing how researchers exploited weaknesses in the card's cryptographic protocol to recover secret keys and clone cards. By reverse engineering the Crypto1 cipher and using known vulnerabilities, the authors show how standard MIFARE Classic cards can be compromised using low cost hardware in real world scenarios. Their findings reveal risks for systems relying on these cards for secure access control and payments.

How is this paper relevant to solving your project problem or addressing your research question? (maximum 100 words)

This work provides background on MIFARE Classic security flaws, motivating the need to investigate additional vulnerabilities, such as side channel leakage. Understanding protocol level attacks helped me design experiments targeting physical signals during authentication and ensured my approach addresses multiple risk factors, not just cryptanalysis.

What are the strengths and weaknesses of the solutions/methods/technologies proposed in this paper? (maximum 100 words)

Strengths:

- Demonstrates a real, repeatable attack on commercial cards
- Offers detailed procedural insights and hardware requirements
- Makes clear the practical risk for widespread deployments

Weaknesses:

- Focuses only on protocol/cipher weaknesses, not physical side channel leakage
- Findings may not generalise to newer or more secure platforms
- Does not propose direct countermeasures for discovered vulnerabilities